# msvtx_el4ecal

MC vs data · overview · rank 39 / 98



rank 39 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    summedCaloClusters/el4ecal
    MS1Vtx_l4ecal_raw
    MS1Vtx_barrel_l4ecal
    MS1Vtx_endcap_l4ecal

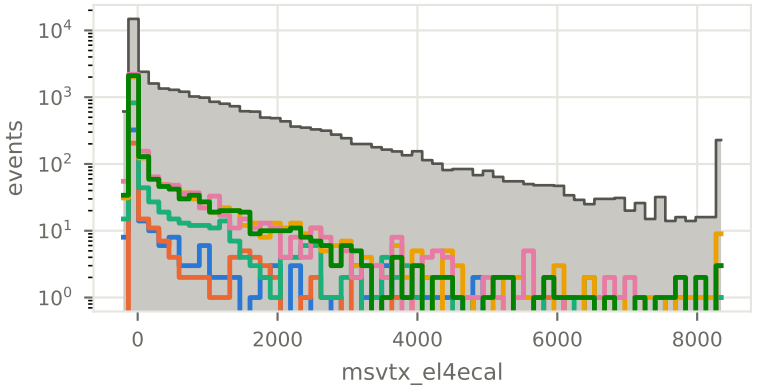
| region | D_equal | noise | D_pooled | 125/15 | 125/55 | 200/50 | 400/100 | 600/150 | 1000/275 | n_bkg | rank |
|---|---|---|---|---|---|---|---|---|---|---|---|
| endcap | 0.337 | 0.022 | 0.340 | 0.368 | 0.293 | 0.342 | 0.331 | 0.345 | 0.348 | 34,970 | 39/98 |
| barrel | 0.243 | 0.051 | 0.251 | 0.272 | 0.201 | 0.222 | 0.266 | 0.266 | 0.237 | 1,089 | 52/98 |
| all | 0.349 | 0.018 | 0.355 | 0.379 | 0.309 | 0.345 | 0.352 | 0.366 | 0.358 | 36,059 | 38/98 |

# msvtx_el4ecal

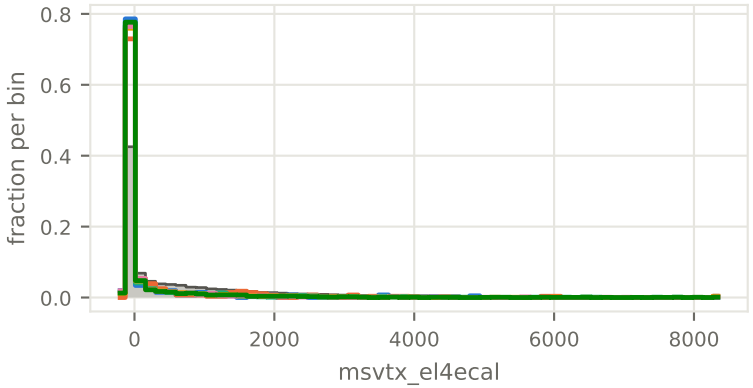
region: ENDCAP · rank 39 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.337 (noise 0.022) D_pooled 0.340 min 0.293 (mH125_mS55) max 0.368 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

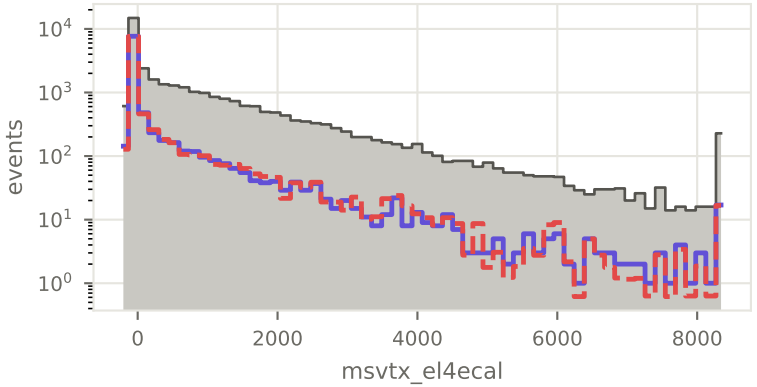



1b · each mass point vs background · normalized

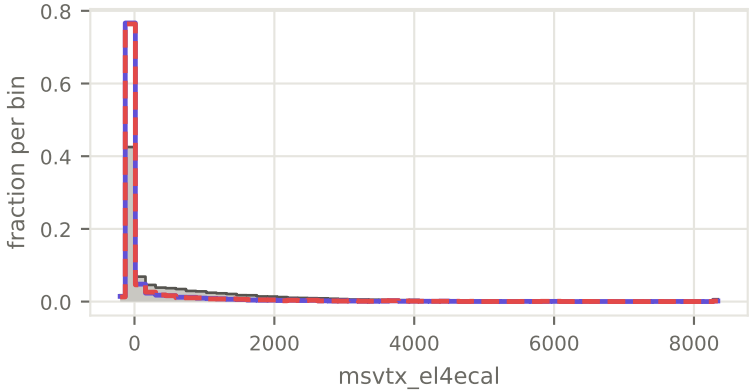



1 · each mass point
background n 34,970
mH125_mS15 D 0.368 n 411
mH125_mS55 D 0.293 n 281
mH200_mS50 D 0.342 n 1,071
mH400_mS100 D 0.331 n 2,648
mH600_mS150 D 0.345 n 2,882
mH1000_mS275 D 0.348 n 2,703

2a · combined signal vs background · raw

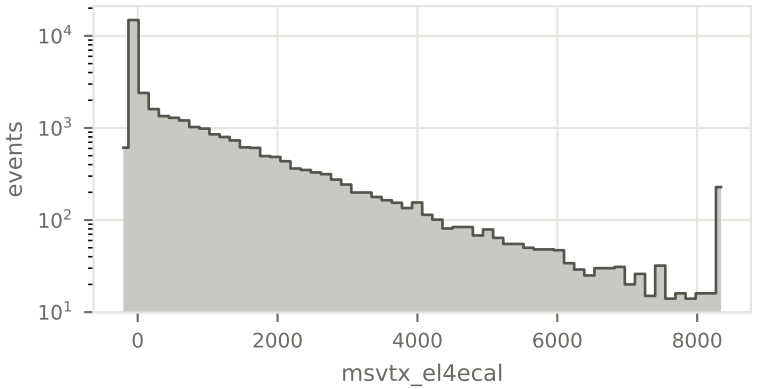



2b · combined signal vs background · normalized

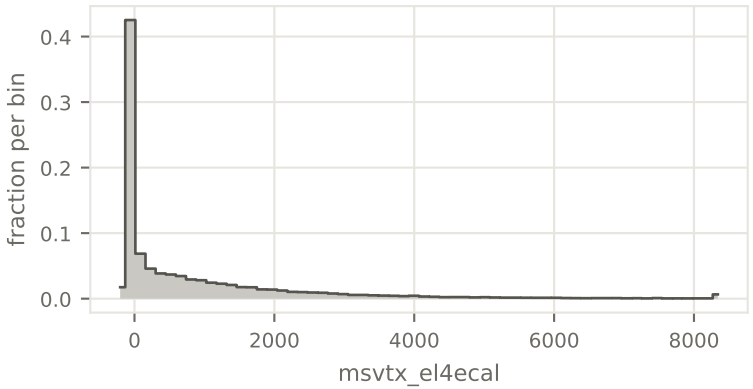



2 · combined signal
background n 34,970
combined, pooled D 0.340 n 9,996
equal weight per mass D 0.337

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

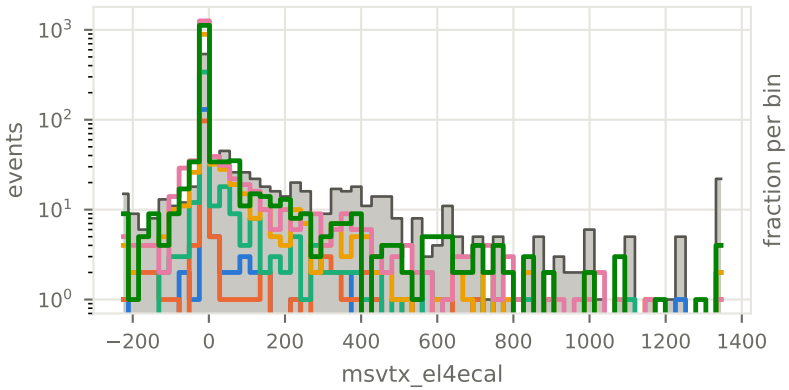
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el4ecal

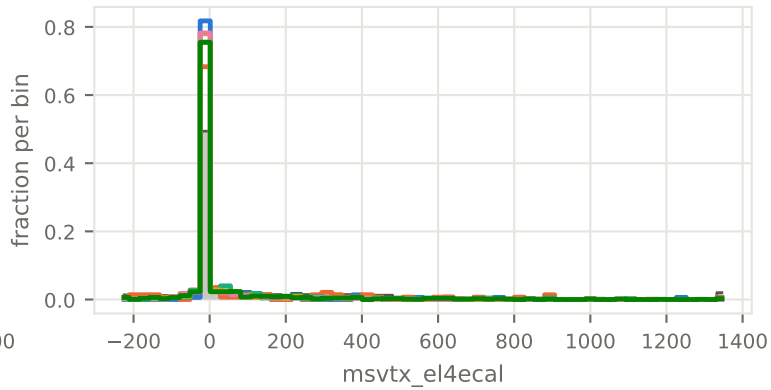
region: BARREL · rank 39 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.243 (noise 0.051) D_pooled 0.251 min 0.201 (mH125_mS55) max 0.272 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

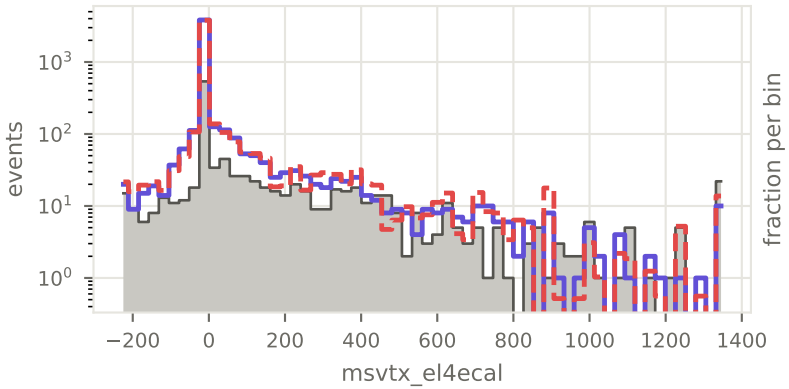



1b · each mass point vs background · normalized

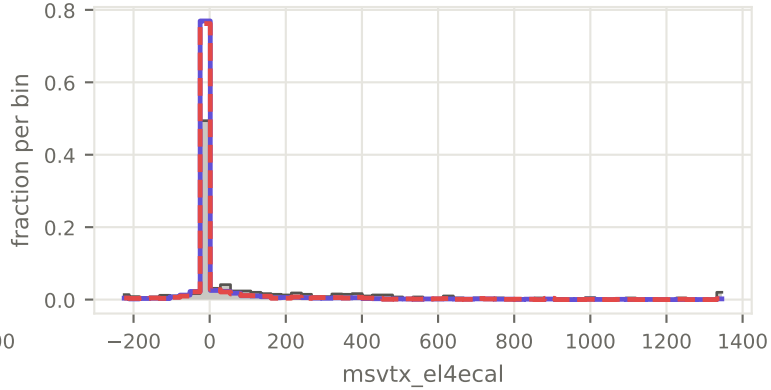



1 · each mass point
background n 1,089
mH125_mS15 D 0.272 n 159
mH125_mS55 D 0.201 n 142
mH200_mS50 D 0.222 n 449
mH400_mS100 D 0.266 n 1,144
mH600_mS150 D 0.266 n 1,608
mH1000_mS275 D 0.237 n 1,485

2a · combined signal vs background · raw

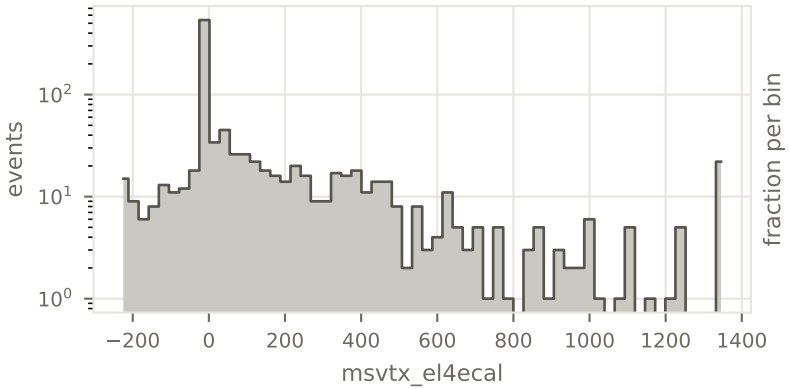



2b · combined signal vs background · normalized

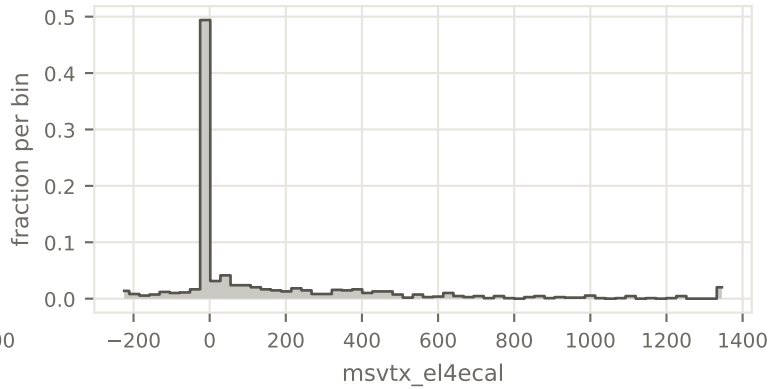



2 · combined signal
background n 1,089
combined, pooled D 0.251 n 4,987
equal weight per mass D 0.243

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

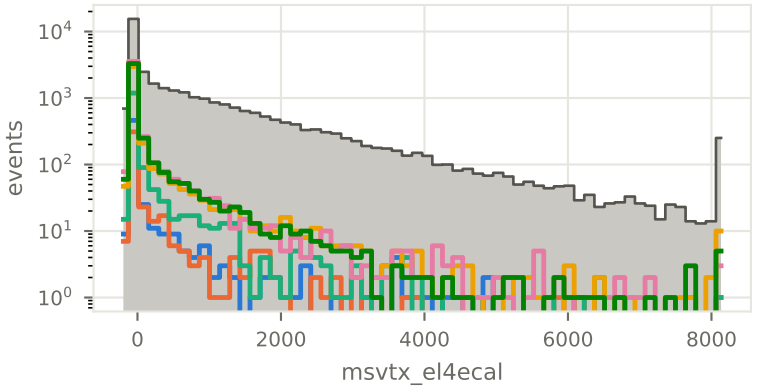
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_el4ecal

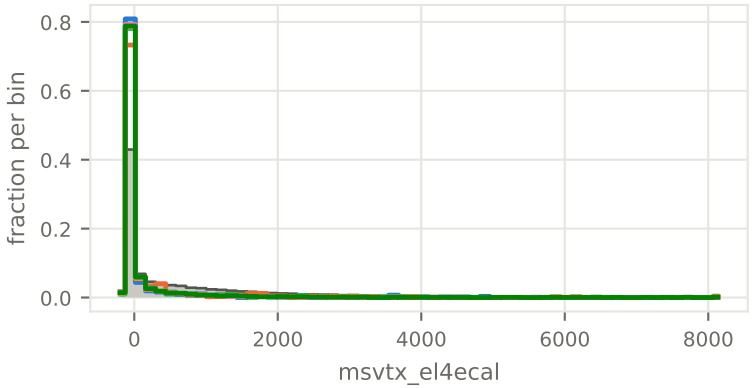
region: ALL · rank 39 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.349 (noise 0.018)  D_pooled 0.355  min 0.309 (mH125_mS55)  max 0.379 (mH125_mS15)  bkg n = 36,059  sig n = 14,983

1a · each mass point vs background · raw

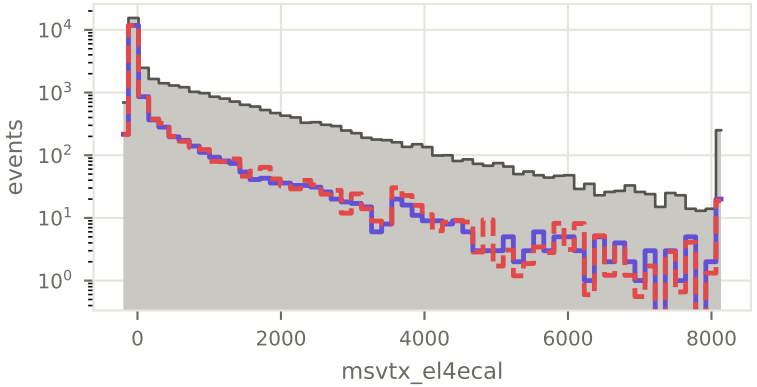



1b · each mass point vs background · normalized

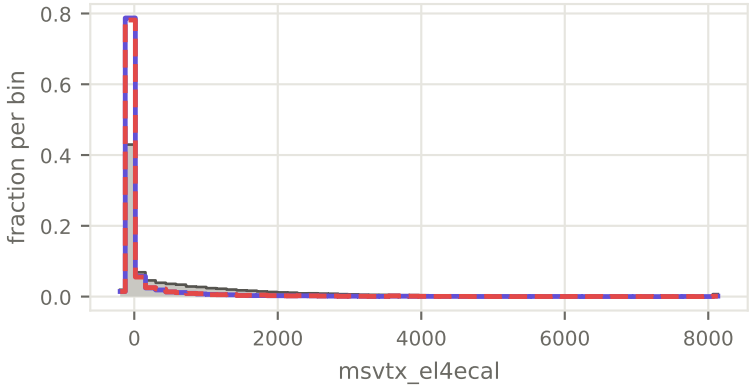



1 · each mass point
background n 36,059
mH125_mS15 D 0.379 n 570
mH125_mS55 D 0.309 n 423
mH200_mS50 D 0.345 n 1,520
mH400_mS100 D 0.352 n 3,792
mH600_mS150 D 0.366 n 4,490
mH1000_mS275 D 0.358 n 4,188

2a · combined signal vs background · raw

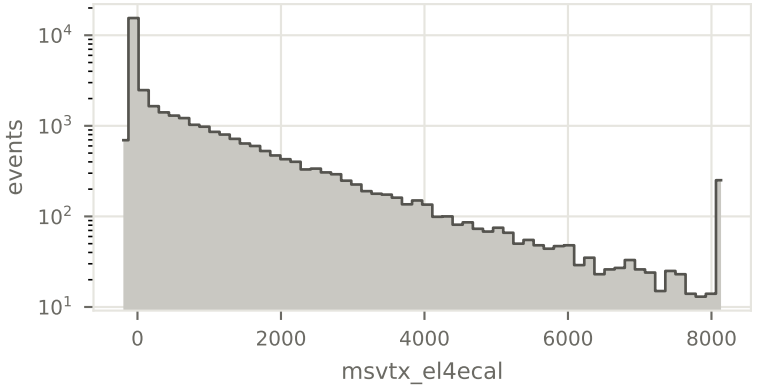



2b · combined signal vs background · normalized

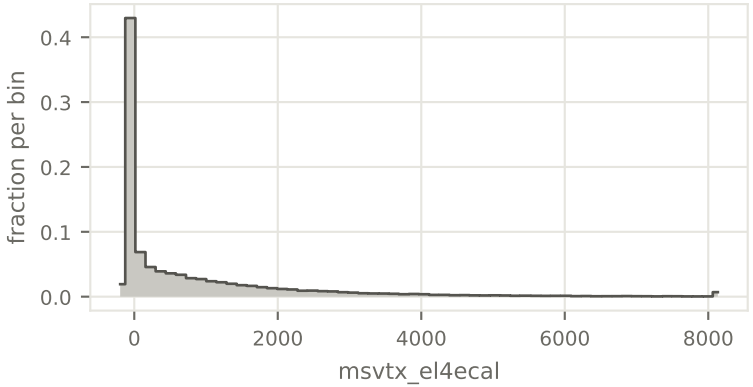



2 · combined signal
background n 36,059
combined, pooled D 0.355 n 14,983
equal weight per mass D 0.349

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
background n 36,059

a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed